

1 Forecast-available signals 2 Leakage-controlled panel 3 Prospective evaluation 4 Category-calibrated alerts

311 history
Calendar
Feature-week weather
Service/geographic IDs



NTA x Week x Category
History at t -> Event at $t+1$
shifted temporal features

$t \rightarrow t+1$ design



LightGBM + XGBoost
Train 2015-22 | Val 2023
Test 2024-25



Ensemble score ->
service threshold -> risk alert
SHAP-assisted review

validation only

Separated retrospective branch

OSM + PLUTO 2026



Retrospective
context check



Not used for
prospective alerts